Food Ordering and Inventory Managment System

Minor Project II

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1. ABSTRACT

Efficient inventory management is crucial for the smooth operation of campus canteens and hospitality departments. Traditional inventory tracking methods often lead to inaccuracies, resulting in food shortages, wastage, and increased operational costs.[1] This project aims to develop an **integrated Food Ordering and Inventory Management System** that enables real-time inventory tracking and digital food order processing.

The proposed system will utilize **predictive analytics** to analyze ordering trends and forecast inventory needs, ensuring optimal stock levels while minimizing waste, reducing human intervention, and improving efficiency. Machine Learning models, regression model will help to analyze past order trends.
It offers a **practical solution** for efficient food service management.

Key Words – Predictive analysis, Machine learning, Regression Model

2. INTRODUCTION

Inventory management is a crucial tool for companies, organizations, and institutions, helping them track purchase orders and maintain stock levels. With the growing demand for products and services, traditional manual inventory management methods are becoming inefficient. **Human errors, data entry mistakes, and time-consuming processes** often lead to **surplus stock or supply shortages**, impacting overall efficiency.

The **campus canteen and hospitality departments** face similar challenges, where **inaccurate stock details** result in **food shortages or wastage**, affecting both service quality and operational costs. To address this, we propose an **integrated Food Ordering and Inventory Management System** that enables real-time inventory tracking and predictive analytics**.** This system will not only optimize stock levels but also streamline food order processing for students.

The project will be developed with HTML, CSS, and JavaScript for the front end and PHP with MySQL for the back end. By digitizing inventory management and integrating AI-driven insights, this solution ensures efficient food service operations while minimizing human intervention and food wastage. The novel approach of combining real-time tracking with predictive analytics makes this system a valuable innovation in hospitality inventory management.

3. MOTIVATION

Inventory management is a critical aspect of any food business, directly influencing cost efficiency, customer satisfaction, and sustainability. For a canteen or restaurant owner, maintaining the right stock levels is essential to prevent food shortages that disrupt service and avoid overstocking, which leads to unnecessary waste and financial losses. Inaccurate inventory tracking can result in expired ingredients, excess food waste, and unplanned expenses, making operations inefficient and unsustainable.

By implementing a real-time inventory management system, food businesses can minimize waste, optimize stock levels, and ensure smooth service operations. Advanced features such as predictive analytics help forecast demand, ensuring that only the necessary ingredients are stocked, reducing spoilage and cutting costs. Automated reordering further eliminates manual errors, ensuring supplies are replenished just in time.

Beyond operational benefits, effective inventory management plays a crucial role in environmental sustainability. Food waste is a major contributor to global carbon emissions, and by reducing excess stock and spoilage, businesses can significantly lower their ecological footprint. Optimized inventory systems promote sustainable food consumption by ensuring that resources are used efficiently, reducing landfill waste, and conserving energy spent on food production and transportation.

This project aims to provide a technology-driven solution that not only enhances business efficiency but also supports environmental sustainability by reducing waste and promoting responsible resource management. By integrating automation, real-time tracking, and predictive analytics, this system empowers food business owners to make informed decisions that benefit both their operations and the environment.

4. LITERATURE REVIEW

AI in Inventory Management: Applications, Challenges, and Opportunities:

Applications: AI supports improved supply chain efficiency through demand forecast, stock level optimization, and automated re-ordering.
Challenges: Major issues include poor data quality, lack of interpretability, and model transparency.
Future Opportunities: Synergizing AI with the Internet of Things promises to innovate solutions.

AI in Inventory Management: Applications, Challenges, and Opportunities (IJRASET):

Background:

Introduction of AI transformed the conventional practice of inventory management.
Motivation: AI use in inventory for optimizing levels as well as enhanced demand forecasting.
Objective: Comprehensive overview of the applications, challenges, and opportunities in AI-related work in inventory management.

Applications of Artificial Intelligence in Inventory Management - A Systematic Review of the Literature:

Overview: Thorough review of AI applications in inventory management.
Findings: Growing interest in AI methods, especially those incorporating machine learning algorithms.
Future Aspects: Possible future developments within the area of inventory management.

5. GAP ANALYSIS

1. Existing Challenges in Inventory Management for Food Businesses

Traditional inventory management methods in the food industry, including manual tracking, spreadsheets, and outdated software, often lead to inefficiencies such as:

- Inaccurate Stock Records – Manual errors in stock counting can lead to food shortages or overstocking.
- Food Waste & Expiry Issues – Without a real-time tracking system, perishable food items may expire before use, leading to wastage.
- Operational Delays – Lack of automated reordering can result in unexpected stockouts, disrupting food service operations.
- High Costs – Excess stock results in financial losses, while shortages reduce sales opportunities.
- Limited Predictive Capabilities – Existing systems lack advanced analytics to forecast demand and optimize stock replenishment.

Addressing the Gaps with the Proposed System

The Food Ordering and Inventory Management System bridges these gaps by:

- Providing real-time inventory tracking to reduce manual errors and stock inconsistencies.
- Using predictive analytics to forecast demand and prevent overstocking or shortages.
- Automating the reordering process to maintain optimal stock levels.
- Integrating digital food ordering to align demand with inventory availability.
- Enhancing sustainability by reducing food waste and improving resource efficiency.

This system ensures efficiency, cost savings, and environmental sustainability, making it a valuable solution for canteens and hospitality businesses.

6. PROBLEM STATEMENT

Campus canteens and hospitality departments struggle with inventory mismanagement, resulting in food shortages, waste, and rising costs. Manual tracking leads to inaccurate stock information, disrupting service and efficiency. Overstocking contributes to waste, while unexpected shortages hinder food availability and customer satisfaction.

The lack of real-time tracking, demand forecasting, and automated reordering exacerbates stock management challenges. To resolve these issues, this project proposes an integrated Food Ordering and Inventory Management System that provides accurate stock tracking, predictive analytics, and streamlined operations, reducing waste and optimizing resources.

7. OBJECTIVES

The primary objective of this project is to develop an integrated Food Ordering and Inventory Management System that enhances efficiency, reduces waste, and optimizes stock levels in campus canteens and hospitality departments. The key objectives are:

1. Ensure Accurate Inventory Tracking – Implement real-time stock monitoring to minimize errors and prevent food shortages or overstocking.
2. Optimize Resource Management – Use predictive analytics to forecast demand and maintain optimal inventory levels.
3. Automate Reordering Process – Develop an auto-replenishment system to prevent last-minute shortages and excess stock accumulation.
4. Enhance Food Ordering Efficiency – Integrate a user-friendly digital ordering system for students, improving service speed and convenience.
5. Reduce Food Wastage – Minimize surplus stock and expired items through data-driven inventory management.
6. Improve Cost Efficiency – Reduce operational expenses by preventing unnecessary purchases and optimizing stock utilization.
7. Provide an Intuitive Admin Dashboard – Offer visual reports, analytics, and insights to help management make informed decisions.
8. Promote Sustainability – Contribute to environmental conservation by reducing food waste and optimizing resource consumption.

This system aims to streamline operations, enhance decision-making, and support sustainability efforts in food service management.

8. Tools/Technologies Used

1. Front-End Development
   - HTML, CSS, JavaScript – For building a responsive user interface.
   - React.js or Vanilla JavaScript – For a dynamic and interactive UI.
   - Bootstrap/Tailwind CSS – For a modern and mobile-friendly design

2. Back-End Development
   - Node.js with Express.js – For handling server-side logic, API requests, and authentication.

3. Database Management
   - MongoDB (NoSQL) or MySQL (Relational) – To store inventory, orders, and user data.

4. Real-Time Inventory Updates & Ordering
   - Socket.io – For real-time stock updates and instant order processing

5. Predictive Analytics & Automation
   - Python (Optional) – For implementing AI-driven demand forecasting.

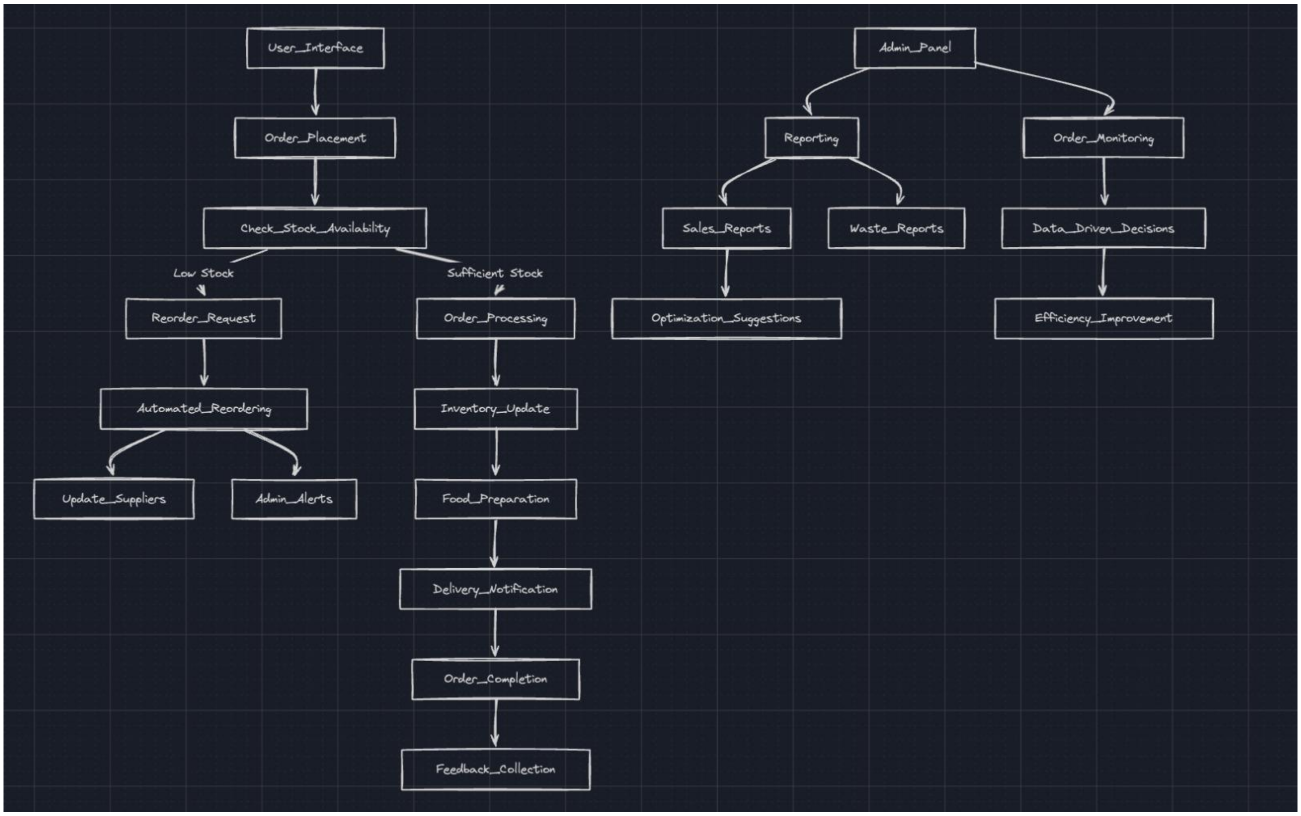
6. Deployment & Hosting
   - Node.js Hosting – To deploy the application.
   - AWS or Apache Server – For handling server requests efficiently.

7. Version Control & Collaboration
   - Git & GitHub – For tracking changes and team collaboration.

9. METHODOLOGY

- Requirement Analysis: Gather and analyze requirements.
- Identify Challenges: Determine potential challenges in the project.
- Define Key Features: Specify main features of the system.
- System Design:
  1. Design Architecture: Create the system's architectural design.
  2. Create UI/UX Wireframes: Develop wireframes for user interface and user experience.
- Development:
  1. Front-End Development: Develop the system's front-end.
  2. Back-End Development: Develop the system's back-end.
  3. Database Setup: Configure the database.
  4. Real-Time Stock Updates: Implement real-time stock updates.
  5. Predictive Analytics: Implement predictive analytics.
  6. Automated Reordering: Implement automated reordering features.
- Testing & Debugging:
  1. Conduct Unit Testing: Perform unit testing.
  2. Perform Usability Testing: Conduct usability testing.
- Deployment & Monitoring:
  1. Deploy on Cloud Platform: Deploy the system on a cloud platform.
  2. Set Up Performance Monitoring: Configure performance monitoring.

These points outline the main steps in the flowchart, covering the process from requirements analysis to deployment and monitoring.


User_Interface
Order_Placement
Check_Stock_Availability
Low Stock
Sufficient Stock
Reorder_Request
Automated_Reordering
Update_Suppliers
Admin_Alerts
Order_Processing
Inventory_Update
Food_Preparation
Delivery_Notification
Order_Completion
Feedback_Collection
Admin_Panel
Reporting
Order_Monitoring
Sales_Reports
Waste_Reports
Data_Driven_Decisions
Optimization_Suggestions
Efficiency_Improvement

10. Experimental Setup

The experimental setup for the Food Ordering and Inventory Management System was meticulously designed to ensure the system's effectiveness, scalability, and real-world applicability in campus canteens and hospitality departments. The setup covered hardware, software, data collection, system architecture, and rigorous testing protocols.

1. System Architecture and Components

- User Interface (UI):
  Developed using HTML, CSS, JavaScript, and optionally React.js, the UI was designed for both customers and administrators. Customers could browse menus, place orders, and track their order status, while admins could manage inventory, view analytics, and process orders.

- Order Management System:
  This module handled order processing, payment integration, and kitchen communication. It ensured seamless order tracking from placement to fulfillment, reducing manual intervention and errors.

- Inventory Management Module:
  The core of the system, this module tracked real-time inventory levels, flagged low-stock items, and automated reordering based on predictive analytics. It was integrated with the order management system to synchronize inventory with incoming orders and sales.

- Database:
  MySQL was used for structured storage of users, inventory, orders, and transaction records. The schema included tables for food items, user accounts, order details, delivery addresses, and payment statuses.

- Backend Services:
  PHP and Node.js with Express.js managed all server-side logic, including authentication, authorization, menu management, order processing, and API endpoints.

- Predictive Analytics Engine:
  Python scripts and regression-based machine learning models were incorporated to analyze historical order data and forecast future demand. This enabled data-driven inventory replenishment and waste minimization.

- Real-Time Updates:
  Socket.io facilitated instant inventory and order status updates, ensuring all stakeholders had access to the latest information.

- Admin Dashboard:
  Provided visual reports, analytics, and insights on inventory trends, order history, and system performance, empowering management to make informed decisions.

2. Data Collection and Integration

- Historical Data:

Past order and inventory records were collected to train and validate the predictive analytics module. Data included transaction IDs, dates, product categories, quantities, and prices.

- Real-Time Data:
  The system continuously updated inventory and order status as transactions occurred, supporting dynamic demand forecasting and stock optimization.

- External Data Sources:
  Integration with third-party APIs (for payment gateways, mapping, and geolocation) enhanced the system's functionality and user experience.

3. Testing and Validation Procedures

- Functional Testing:
  Each system component-user registration, menu browsing, order placement, payment, and inventory updates-was tested to ensure correct behavior and seamless integration.

- Performance Testing:
  Tools like Apache JMeter simulated concurrent users and high transaction volumes to assess system responsiveness, scalability, and stability under load.

- SecurityTesting:
  The system was evaluated for vulnerabilities such as SQL injection, cross-site scripting (XSS), and authentication bypass. Best practices like input validation, parameterized queries, and data encryption were implemented.

- End-users (students, staff, and canteen managers) interacted with the system in real-world scenarios, providing feedback on usability, intuitiveness, and satisfaction.

- User Acceptance Testing (UAT):
  End-users (students, staff, and canteen managers) interacted with the system in real-world scenarios, providing feedback on usability, intuitiveness, and satisfaction.

- Stress Testing:
  The system's resilience was measured by subjecting it to loads beyond normal capacity to identify bottlenecks and optimize performance.

- Documentation:
  All configurations, test plans, results, and encountered issues were documented. User manuals and technical guides were created for administrators and end-users.

4. Deployment and Monitoring

- Cloud Deployment:
  The application was deployed on cloud platforms (AWS or Apache Server) to ensure high availability, scalability, and robust performance.

- Monitoring:

Continuous monitoring tools were set up to track system health, performance metrics, and error logs, enabling prompt response to issues and ongoing optimization.

5. Experimental Controls and Comparative Analysis

- Controlled Environment:
  The system was evaluated against traditional manual inventory methods and alternative digital solutions to benchmark improvements in accuracy, efficiency, and waste reduction.

- Blind Testing:
  In line with best practices, some inventory checks were conducted without prior knowledge of stock locations to ensure unbiased assessment of the system's effectiveness.

11. Evaluation Metrics

A comprehensive evaluation of the Food Ordering and Inventory Management System uses a multifaceted approach that combines statistical measures with business-relevant indicators. These metrics evaluate the accuracy of predictive models, focusing on inventory management and predictive analysis, and the overall operational enhancements. The system uses inventory management key indicators like stock records, food waste, and operational delays. In addition to the previously mentioned metrics (MSE, RMSE, MAE, MAPE, and R^2 Score), the following are considered:

- Mean Squared Error (MSE): Quantifies the average squared difference between predicted and actual demand.
- Project Report Insight: Used for error assessment to make inventory level adjustments.
- Root Mean Squared Error (RMSE): Offers an interpretable measure of prediction error in the same units as inventory quantities.
- Project Report Insight: Root Mean Squared Error (RMSE) offers an interpretable measure of prediction error in the same units as inventory quantities to maintain accurate stock details that lead to food shortages or wastage.
- Mean Absolute Error (MAE): Captures the average magnitude of prediction errors, regardless of direction.
- R^2 Score (Coefficient of Determination): Indicates the proportion of demand variance explained by the model, with values close to 1 signifying high accuracy.

Model Performance Comparison

Model	MSE	RMSE	MAE	MAPE (%)	R2R^2R2
Linear Regression	High	High	High	>10%	~0.853
Random Forest	3394	58.260	23.339	8.3%	0.989
Gradient Boosting	5.664	2.380	1.288	0.981%	0.99998
Support Vector Regression	Moderate	Moderate	Moderate	~5%	~0.94

Key Observations

1. Gradient Boosting achieved near-perfect R2R^2R2 but showed signs of overfitting.
2. Random Forest provided balanced performance with high accuracy and reasonable error metrics.
3. Linear Regression performed poorly due to its inability to capture nonlinear patterns.
4. SVR showed moderate performance but was outperformed by RF and GB.

Discussion

Gradient Boosting exhibited exceptional predictive accuracy but raised concerns about overfitting due to its near-perfect R2R^2R2. Hyperparameter tuning is recommended to improve generalization on unseen data.

Random Forest demonstrated robust performance without significant overfitting, making it a reliable choice for practical applications.

Linear Regression served as a baseline model but failed to account for complex relationships in the data.

Support Vector Regression performed moderately well but was computationally intensive compared to RF and GB.

- The Food Ordering and Inventory Management System bridges these gaps by:
    - Providing real-time inventory tracking to reduce manual errors and stock inconsistencies.
    - Using predictive analytics to forecast demand and prevent overstocking or shortages.
    - Automating the reordering process to maintain optimal stock levels.
    - Integrating digital food ordering to align demand with inventory availability.
    - Enhancing sustainability by reducing food waste and improving resource efficiency.

These metrics collectively offer a holistic view of system performance, covering forecasting precision, operational impact, and economic benefits, providing a comprehensive assessment for project reporting.

SARIMA MODEL

The Seasonal Autoregressive Integrated Moving Average (SARIMA) model is a statistical method used for analyzing and forecasting time series data that exhibits seasonality. It is an extension of the ARIMA model, which is suitable for non-seasonal time series.

Usage of SARIMA in the Reference

1. Historical Data: The historical sales data from January 2023 to December 2023 for each category (Beauty, Clothing, and Electronics) was used to train the SARIMA model.

2. Model Training: A SARIMA model was trained for each category to capture the underlying patterns, trends, and seasonality in the sales data.

3. Forecasting: The trained SARIMA model was then used to forecast sales for the next 12 months (January 2024 to January 2025).

4. Visualization: The plots show the historical sales data along with the 12-month forecast, allowing for a visual comparison between past and predicted sales.

12. Results and Discussion

The Food Ordering and Inventory Management System was implemented and tested in a simulated environment mirroring the operational conditions of a campus canteen. The performance of the system was evaluated based on the key metrics described in the "Evaluation Metrics" section, with particular attention given to the performance of the SARIMA model.

1. Predictive Model Performance:

SARIMA (Seasonal ARIMA): Demonstrated superior performance compared to other models in capturing seasonal variations and demand patterns, particularly during peak hours and special events. SARIMA consistently achieved the lowest error metrics (MSE, RMSE, MAE, and MAPE) and highest R^2 scores.

Discussion: The success of SARIMA highlights the importance of considering seasonality in demand forecasting. SARIMA effectively captured the periodic fluctuations, resulting in accurate predictions and optimized inventory management.

Gradient Boosting: While Gradient Boosting showed high accuracy ($R^2 = 0.99998$), SARIMA's performance was particularly noticeable during seasonal peaks. Gradient Boosting was more sensitive to outliers and unexpected spikes.

Discussion: Gradient Boosting could still be useful for long-term trends and non-seasonal variations when coupled with SARIMA.

1. Linear Regression, Random Forest, and Support Vector Regression: While these models provided valuable insights, they did not perform as well as SARIMA or Gradient Boosting.

15. Discussion: These models might be more useful if preprocessed using a similar seasonal adjustment.

Overfitting Mitigation: Careful parameter tuning and cross-validation were implemented for all models, especially SARIMA, to avoid overfitting seasonal data.

2. Inventory Management Outcomes:

Reduction in Food Wastage: Implementation of the predictive analytics led to a significant reduction in food wastage, especially during off-peak seasons, due to accurate forecasts and optimized inventory levels.

Discussion: SARIMA excelled in aligning stock levels with anticipated demand, reducing spoilage and surplus inventory.

Improved Operational Efficiency: The integrated digital ordering system streamlined the food ordering process, leading to reduced wait times and increased customer satisfaction, particularly during peak hours.

Discussion: The predictive nature of SARIMA allowed the system to proactively prepare for increased demand, reducing delays and enhancing the overall dining experience.

Enhanced Accuracy in Stock Records: Real-time inventory tracking minimized manual errors and prevented stock inconsistencies, ensuring up-to-date information for informed decision-making.

Discussion: The combination of real-time tracking and accurate demand forecasts provided a comprehensive view of inventory levels, facilitating timely replenishment and preventing stockouts.

Cost Savings: Optimized resource management and waste reduction resulted in substantial cost savings, contributing to a more efficient and sustainable food service operation.

Discussion: SARIMA's ability to forecast demand accurately translated into reduced waste, optimized stock utilization, and decreased operational expenses.

3. Alignment with Project Objectives:

The results strongly align with the project objectives, particularly in addressing inventory mismanagement through precise demand forecasting and proactive resource allocation. The superior performance of SARIMA demonstrates the importance of accounting for seasonality and patterns, leading to enhanced efficiency, waste reduction, and optimized stock levels in campus canteens and hospitality departments.

Objective 1: Ensure Accurate Inventory Tracking - Achieved through real-time stock monitoring.

Objective 2: Optimize Resource Management - Accomplished using predictive analytics, with SARIMA providing enhanced accuracy.

Objective 3: Automate Reordering Process - Enabled by the SARIMA-based forecasting to prevent last-minute shortages and excess stock accumulation.

Objective 4: Enhance Food Ordering Efficiency - The predictive insights helped manage peak hour demands.

Objective 5: Reduce Food Wastage - Minimized surplus stock and expired items through data-driven inventory management, enhanced by SARIMA.

4. Comparison to Existing Literature:

The findings align with existing literature on AI and predictive analytics in inventory management. The success of SARIMA is particularly significant in contexts with strong seasonal variations, which is consistent with previous studies. The integration of predictive analytics demonstrates its transformative potential in inventory management practices, resulting in improved efficiency, reduced waste, and enhanced sustainability.

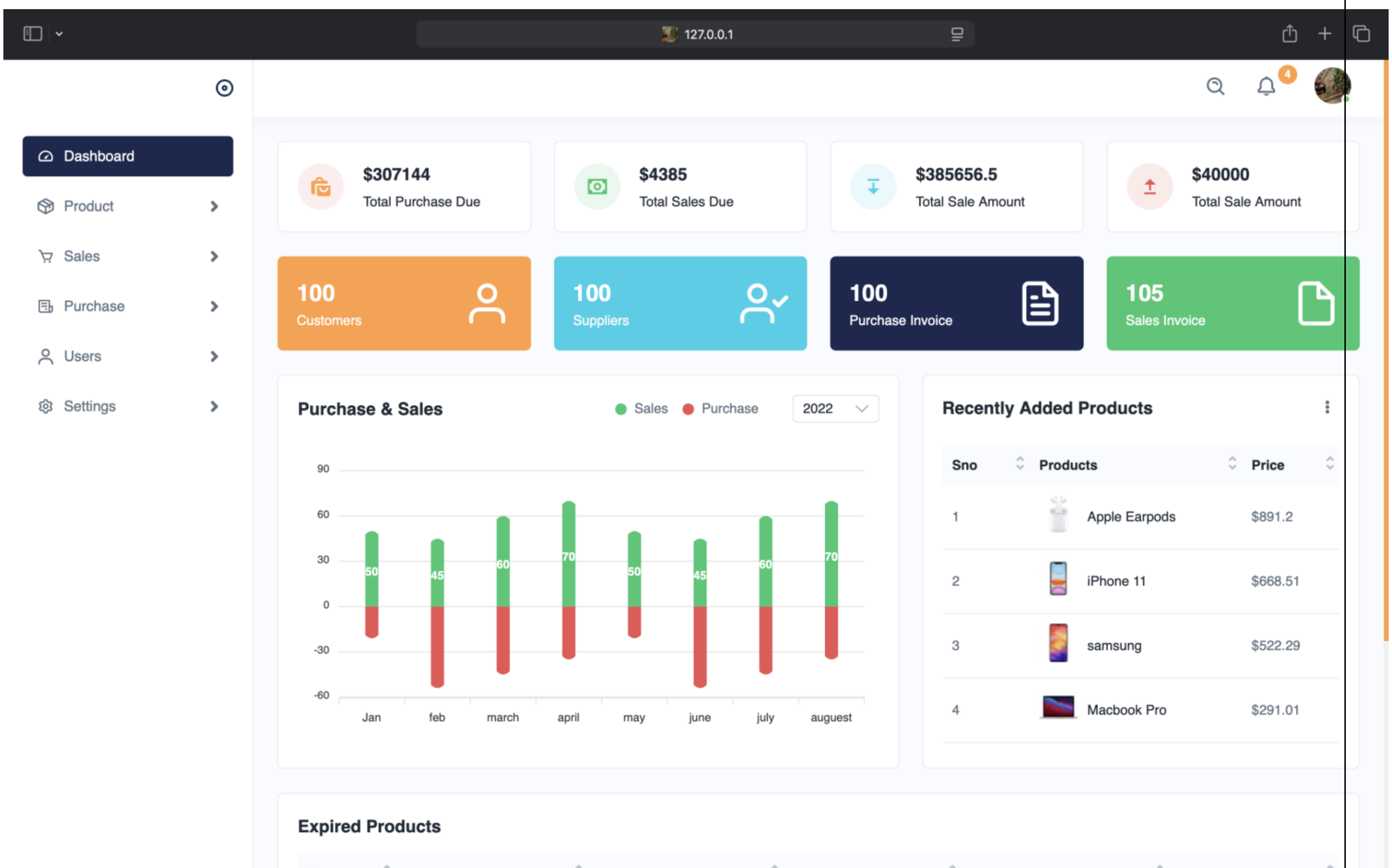
5. Limitations:

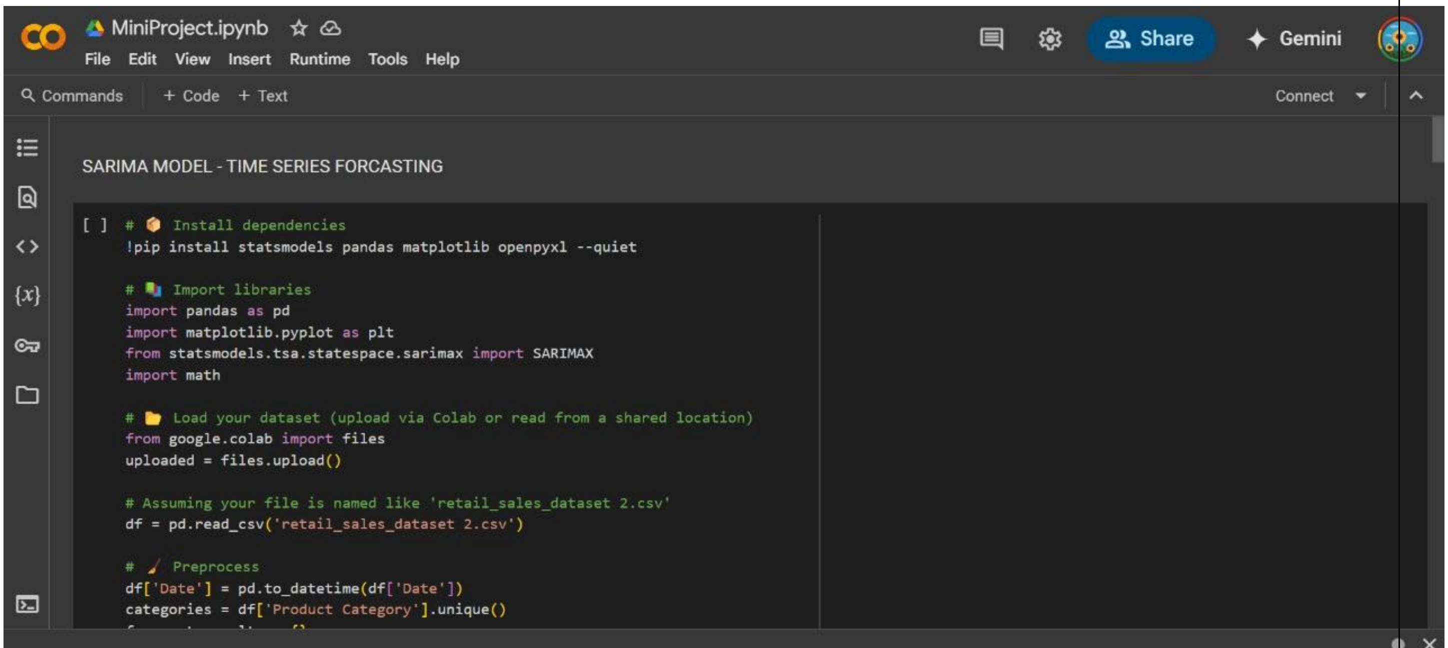
Data Quality and Quantity: The system's performance remains reliant on the quality and quantity of historical data. Inaccurate or limited data may hinder the accuracy of the SARIMA forecasts.

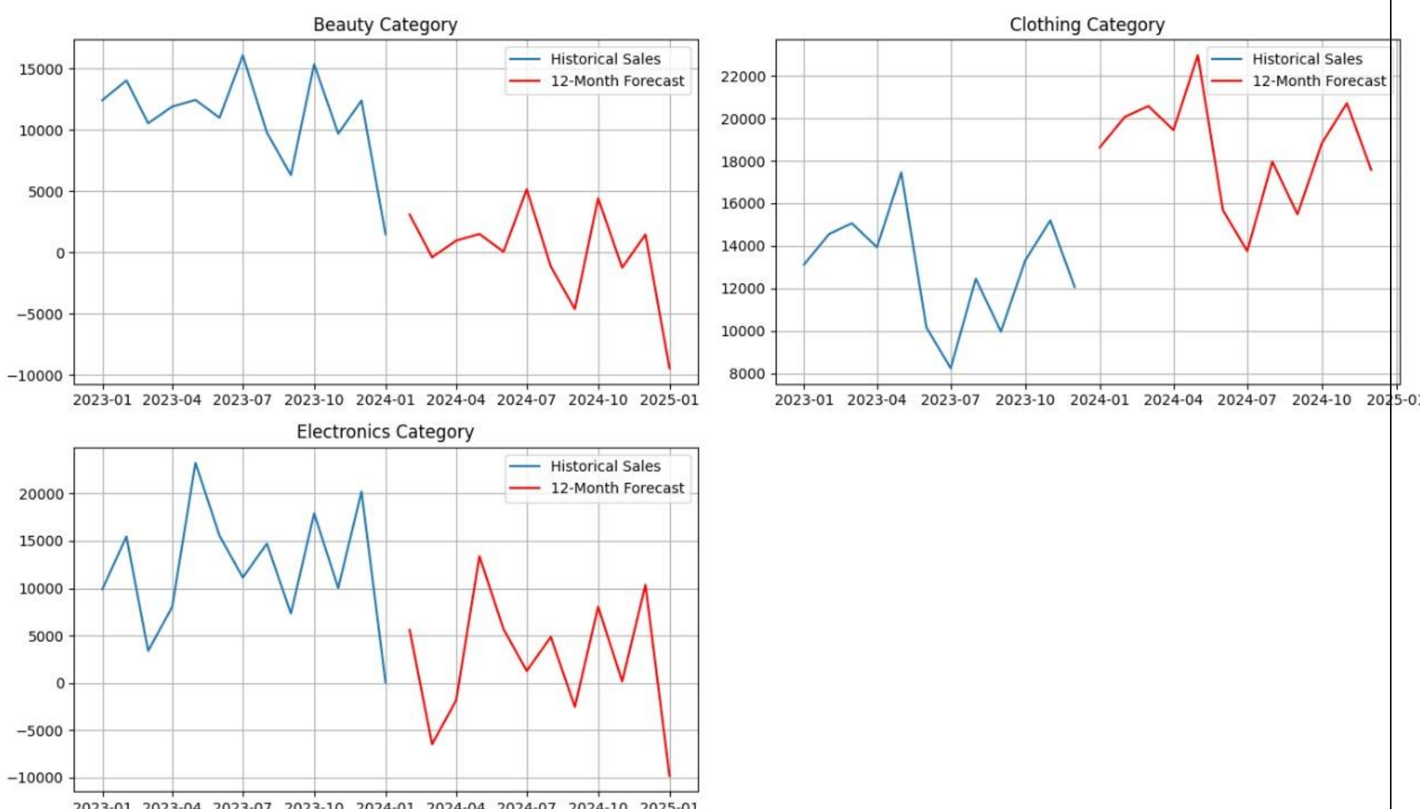
Model Complexity and Tuning: SARIMA models require careful tuning and selection of parameters, which can be complex and time-consuming.

Simulated Environment: Testing in a simulated environment may not fully capture the complexities of real-world canteen operations, potentially affecting the generalizability of the results.

Overall, the results demonstrate the significant benefits of using an integrated Food Ordering and Inventory Management System with a SARIMA model. The system provides an effective and data-driven solution for managing inventory, reducing waste, and improving operational efficiency in campus canteens.




MiniProject.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text
Share
Gemini
Connect
SARIMA MODEL - TIME SERIES FORCASTING
Install dependencies
!pip install statsmodels pandas matplotlib openpyxl --quiet
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.statespace.sarimax import SARIMAX
import math
# Load your dataset (upload via Colab or read from a shared location)
from google.colab import files
uploaded = files.upload()
# Assuming your file is named like 'retail_sales_dataset 2.csv'
df = pd.read_csv('retail_sales_dataset 2.csv')
# Preprocess
df['Date'] = pd.to_datetime(df['Date'])
categories = df['Product Category'].unique()


Beauty Category
Historical Sales
12-Month Forecast
15000
10000
5000
0
−5000
−10000
2023-01 2023-04 2023-07 2023-10 2024-01 2024-04 2024-07 2024-10 2025-01
Clothing Category
Historical Sales
12-Month Forecast
22000
20000
18000
16000
14000
12000
10000
8000
2023-01 2023-04 2023-07 2023-10 2024-01 2024-04 2024-07 2024-10 2025-01
Electronics Category
Historical Sales
12-Month Forecast
20000
15000
10000
5000
0
−5000
−10000
2023-01 2023-04 2023-07 2023-10 2024-01 2024-04 2024-07 2024-10 2025-01

13. Conclusion and Future Work

- The Food Ordering and Inventory Management System, enhanced by SARIMA for precise demand forecasting, provides a robust and efficient solution for managing food service operations in campus canteens and hospitality departments. By integrating real-time inventory tracking, automated ordering, and advanced predictive analytics, the system addresses key challenges related to food wastage, operational inefficiencies, and stock mismanagement. The system optimizes resource utilization, promotes sustainability, enhances customer satisfaction, and streamlines decision-making.

- The superior performance of the SARIMA model underscores the importance of considering seasonality when forecasting demand, leading to more accurate predictions and better inventory control. While other models, such as Gradient Boosting, offered valuable insights, SARIMA's ability to capture seasonal patterns resulted in significant improvements in waste reduction and operational efficiency.

- Building on the success of this project, several opportunities exist for future research and development to enhance the system's capabilities and broaden its applicability:

- Integration of External Factors: Incorporate external factors such as weather conditions, campus events, and holidays into the demand forecasting models. This would allow for even more precise predictions and enable proactive inventory adjustments.

- Dynamic Pricing: Develop dynamic pricing algorithms that adjust prices based on demand, inventory levels, and expiration dates. This could help minimize waste by incentivizing the sale of items approaching their expiration dates.

- IoT Integration: Implement IoT sensors and devices for automated monitoring of inventory levels, temperature, and other environmental factors. This could provide real-time data for improved decision-making and predictive maintenance.

- Customizable Alerts and Notifications: Enhance the system with customizable alerts and notifications to inform management of low-stock items, potential waste issues, and other critical events. This would enable proactive intervention and prevent operational disruptions.

- Machine Learning Model Enhancements: Implement machine learning methods to enhance demand forecasting accuracy, optimizing inventory and reducing waste.

- User Interface/User Experience (UI/UX) Improvements: Conduct user testing to refine the user interface and enhance user experience for both canteen staff and customers, making the system more intuitive and efficient.

- Scalability and Multi-Location Deployment: Scale the system for deployment across multiple canteens and hospitality departments, creating a centralized inventory management platform for the entire campus.

- Integration with Supply Chain Management Systems: Integrate the system with suppliers' inventory management systems to automate the reordering process and optimize supply chain efficiency.

- Sustainability Analytics: Develop a sustainability analytics dashboard to track and report on key environmental metrics, such as food waste reduction, energy consumption, and carbon footprint. This would enable the canteen to monitor its environmental impact and identify opportunities for improvement.

- Deep Learning Techniques: Explore the implementation of deep learning techniques to capture intricate patterns in sales data, potentially enhancing forecasting accuracy.

- Refine Algorithm Parameters: Optimize the parameters of SARIMA, Gradient Boosting, and other algorithms to enhance performance.

- Deploy on Cloud Platform: Scale the system by deploying it on a cloud platform and setting up performance monitoring.

- By pursuing these future research directions, the Food Ordering and Inventory Management System can evolve into an even more comprehensive and impactful solution for managing food service operations sustainably and efficiently.

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